

To appear in the *Proceedings of the 1996 Barrett Lectures*.

ACCURATE STURM-LIOUVILLE EIGENFUNCTION COMPUTATION IN MATHEMATICAL SOFTWARE

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Abstract

In the past few years several high quality Sturm-Liouville software packages have been written (the NAG routine SL02FM, SLEDGE, SLEIGN). All excel at providing eigenfunction estimates at a finite set of points but only the NAG routine provides interpolation capability, and this by a fairly crude method (especially when compared with the sophistication of its eigenvalue solver). We seek efficient, accurate methods of approximating Sturm-Liouville eigenfunctions based on a small sample of given accurate values. Since the author is most familiar with SLEDGE, it will get most of the attention. The proposed methods are closely tied to SLEDGE's basic approximations, its mesh selection, and its classification scheme, so these are briefly reviewed. The known theory on SLEDGE's approximate eigenfunction errors are also discussed. Several alternatives are studied, both numerically and analytically. The first, coefficient approximation, is just SLEDGE's underlying algorithm. The second is based on residual correction, and the other is a piecewise L-spline of sorts. While we concentrate on regular problems for which the mathematics is easier, we mention extensions to the more interesting singular case.

This research was partially supported by National Science Foundation grant DMS-9322615 to the Colorado School of Mines.

1 INTRODUCTION

In this paper the Sturm-Liouville problem

$$-(p(x)u')' + q(x)u = \lambda r(x)u \quad (1.1)$$

is studied for $a < x < b$; we are especially interested in approximating the eigenfunctions associated with (1.1). The package SLEDGE of Pruess and Fulton (1993) is capable of generating numerical approximations to u and to pu' by a method discussed in [11]. Unfortunately, output approximations are produced only at finite set of values of the independent variable. It is desirable to produce approximations a posteriori at an arbitrary value of the independent variable. The preferred mode of operation for SLEDGE is to have it produce its approximations to $u(x)$ and $(pu')(x)$ at points x in its initial mesh. Since SLEDGE works very efficiently when it is allowed to choose this initial mesh based on heuristics from its classification of the problem [7], we prefer not to modify SLEDGE's basic algorithm, but only to add an interpolation capability. We also mention that the Sturm-Liouville package SLEIGN of Bailey, Gordon, and Shampine (1978), or Bailey, Garbow, Kaper, and Zettl (1991) also produces eigenfunction output only at discrete points specified by the user; the methods discussed in this paper are germane to that code as well. There are three goals of this research:

- being able to plot the $u(x)$ estimate using many x points;
- to retain the accuracy of the mesh point data from SLEDGE at any x in $[a, b]$;
- achieve the correct qualitative behavior, e.g., for Dirichlet boundary conditions the k th eigenfunction should have k interior zeros, and exponentially decaying regions should not have extraneous oscillations.

Note that we don't need much accuracy for the first goal, so the second may be ignored in plotting applications when data are available at tight tolerances. The third goal is nearly always important.

We begin with a review of the basic SLEDGE algorithm including its automatic classification and mesh selection heuristics. The underlying mathematical theory is also presented. Then various possible approaches are suggested for generating approximations to the eigenfunctions based on accurate SLEDGE output at discrete values (the points of SLEDGE's initial mesh). Several numerical illustrations are given which use the three approaches that seem to be especially promising. We conclude with a discussion of future directions of this work.

2 THE BASIC SLEDGE APPROXIMATION

For now assume (1.1) is regular. Given an initial mesh (constructed as discussed below) comprised of N subintervals with endpoints $\{a = x_1 < x_2 < \dots < x_{N+1} = b\}$, on each (x_n, x_{n+1}) let $\hat{u}$ and $\hat{\lambda}$ satisfy the “approximate” differential equation

$$-(\hat{p}\hat{u}')' + \hat{q}\hat{u} = \hat{\lambda}\hat{r}\hat{u} \quad (2.1)$$

subject to the analogous boundary conditions as (1.1).

For the coefficient functions in (2.1), SLEDGE uses step function approximations to the respective coefficients of (1.1). Then (2.1) can be “solved” exactly. Set

$$\tau_n = \frac{\lambda r_n - q_n}{p_n} \quad (2.2)$$

and

$$\omega_n = \sqrt{|\tau_n|}, \quad (2.3)$$

with p_n the value of $\hat{p}$ on the n th piece, etc. With

$$\phi_n(t) = \begin{cases} \frac{\sin \omega_n t}{\omega_n} & \tau_n > \epsilon \\ \frac{\sinh \omega_n t}{\omega_n} & \tau_n < -\epsilon \\ t - \tau_n \frac{t^3}{6} + \tau_n^2 \frac{t^5}{120} & |\tau_n| \leq \epsilon \end{cases} \quad (2.4)$$

for some small positive ϵ ,

$$\phi'_n(t) = \begin{cases} \cos \omega_n t & \tau_n > \epsilon \\ \cosh \omega_n t & \tau_n < -\epsilon \\ 1 - \tau_n \frac{t^2}{2} + \tau_n^2 \frac{t^4}{24} & |\tau_n| \leq \epsilon \end{cases}, \quad (2.5)$$

and we have

$$\hat{u}(x) = \hat{u}_n \phi'_n(x - x_n) + \hat{u}'_n \phi_n(x - x_n)/p_n. \quad (2.6)$$

Note: $\hat{u}_n = \hat{u}(x_n)$ and $\hat{u}'_n = (\hat{p}\hat{u}')(x_n)$. The validity of these expressions depends on the facts that $\phi_n(0) = 0$, $\phi'_n(0) = 1$, and (at least for $|\tau_n| > \epsilon$) $\phi''_n(t) = -\tau_n \phi_n(t)$. SLEDGE actually solves for $\hat{\lambda}$, $\{\hat{u}_n\}$, and $\{\hat{u}'_n\}$. In fact, to achieve high accuracy efficiently, Richardson’s h^2 extrapolation is used over a sequence of meshes to produce output estimates for λ , $\{u_n\}$, and $\{u'_n\}$. Experience has shown that these are nearly always within the user’s requested tolerance.

Initially SLEDGE analyzes the coefficient functions p , q , and r to determine (when possible) the classification of the problem, e.g.,

regular or singular, limit circle or limit point, oscillatory or nonoscillatory. Infinite intervals are implicitly transformed into finite ones by the change of independent variable $x = -1/t$ on the first or last initial mesh interval. For regular singular points, the leading term of the principal solution is found. All this is accomplished, e.g., for $p(x)$, by trying to find E_p and c_p such that

$$p(x) \sim c_p(x - a)^{E_p}$$

near the singular point a . See [7] for details.

The initial heuristics for determining a mesh examine two factors:

- the size of $\|p'\|$, $\|q'\|$, and $\|r'\|$ (max norms) relative to the subinterval widths;
- the sign and magnitude of the Liouville normal form potential $Q(t)$.

The error analysis for $p - \hat{p}$, etc., suggests that the subintervals should be smaller (denser x_n) where $|p'|$ is larger. But on physical grounds, we probably should place many mesh points near local minima of $Q(t)$, at least when it is negative there. We have an iterative heuristic that attempts to distribute points so the product of these two factors is approximately constant over the mesh. The actual number of intervals N is a function of the user's input tolerance, ranging from 9 to 21 as the tolerance varies from 10^{-3} to 10^{-15} , and whether or not the endpoints are singular.

For singular problems, depending on the output from the classification algorithm, more points may be added in the neighborhood of the singular point. The number and distribution is a function of many factors including the requested tolerance; the exact heuristics were developed mainly by experimentation over our large test set.

In carrying out the Richardson's extrapolations, estimates for u_n and u'_n values are saved only at points of the *initial* mesh. Because these are so few in number (for efficiency reasons) there is a need for another algorithm which can take these values and cheaply compute a $\bar{u}(x)$ for any x that retains the accuracy of these SLEDGE values.

For the remainder of the paper we concentrate on one $[x_n, x_{n+1}]$ interval which we prefer to denote by $[\alpha, \beta]$, and assume we have accurate values for λ , $u(\alpha)$, $(pu')(\alpha)$, $u(\beta)$, and $(pu')(\beta)$. All approximations further subdivide this interval into M pieces which we denote by $[\zeta_k, \zeta_{k+1}]$ for $1 \leq k \leq M$. The width of the k th piece is h_k . It is convenient to set

$$W(x) = \lambda r(x) - q(x)$$

with $\widehat{W}(x)$ defined analogously. For simplicity, we assume that $p(x)$ and its first few derivatives are continuous on $[\alpha, \beta]$. It is also assumed

that to the software p' is known, as well as p , q , and r , but no other derivatives are available.

Note: due to SLEDGE's relatively coarse initial meshes, standard piecewise polynomial interpolants have great difficulty with the third goal stated in the introduction; consequently, all the methods we consider are based, at least to some degree, on approximations like (2.6) which is generally qualitatively correct.

3 SLEDGE-LIKE APPROXIMATIONS

This method is based on SLEDGE's algorithm of coefficient approximation applied to uniform subdivisions of $[\alpha, \beta]$, i.e., $h = (\beta - \alpha)/M$. On each subinterval $[\zeta_k, \zeta_{k+1}]$ let $\bar{u}$ be the exact solution of the approximate differential equation

$$-p_k \bar{u}'' + q_k \bar{u} = \lambda r_k \bar{u}. \quad (3.1)$$

Note that this differs from the SLEDGE approximation because we are using the exact λ of (1.1); in contrast (2.1) has this as an unknown as well as the eigenfunction.

Then from (2.4) and (2.5), with the interpolant denoted by $I_1 u$, we have

$$(I_1 u)(x) := \gamma_k \phi_k'(x - \zeta_k) + \delta_k \phi_k(x - \zeta_k)/p_k, \quad (3.2)$$

and the corresponding pu' approximation is

$$(I_1' u)(x) := -p_k \tau_k \gamma_k \phi_k(x - \zeta_k) + \delta_k \phi_k'(x - \zeta_k),$$

where $\gamma_k := (I_1 u)(\zeta_k)$ and $\delta_k := (I_1' u)(\zeta_k)$. We require

$$\gamma_1 = u(\alpha) \quad \text{and} \quad \gamma_{M+1} = u(\beta)$$

as interpolation conditions. Continuity requires

$$\gamma_k \phi_k'(h_k) + \delta_k \phi_k(h_k)/p_k - \gamma_{k+1} = 0$$

and

$$-p_k \tau_k \gamma_k \phi_k(h_k) + \delta_k \phi_k'(h_k) - \delta_{k+1} = 0$$

for $k = 1, 2, \dots, M$. There results a $2M + 2$ by $2M + 2$ linear system for the nodal values $\{\gamma_k\}$ and $\{\delta_k\}$. The other two interpolation conditions on pu' at the ends can be used to provide an indication of accuracy. Alternatively, a step of Richardson's extrapolation can be used to estimate errors. Unfortunately, by not forcing interpolation of pu' at the end points (points in SLEDGE's original mesh), we lose global continuity of $p(I_1 u)'$. Requiring this interpolation leads to an

overdetermined linear system, consistent only when λ and $u(x)$ happen to be exact solutions of the approximate differential equation (2.1). Experience has shown that the resulting $I_1 u$ is somewhat more accurate than the derivative approximation $I_1' u$, but the latter is tolerable, especially for plotting pu' which was one of the above-stated goals of this research.

Since this algorithm is based on the SLEDGE approximation but with a possibly finer grid, convergence results can be obtained from minor modifications of the arguments in the basic reference of Pruess (1973b). To set the notation more precisely, let $\bar{u}$ satisfy

$$-\bar{p}\bar{u}'' + \bar{q}\bar{u} = \bar{\lambda}\bar{r}\bar{u}$$

on $[\alpha, \beta]$ with $\bar{p} = p_k$ on $[\zeta_k, \zeta_{k+1}]$, and similarly for $\bar{q}$ and $\bar{r}$. We assume a classical solution, i.e., $\bar{u}, \bar{p}\bar{u}' \in C[\alpha, \beta]$. The value for $\bar{\lambda}$ is the high accuracy SLEDGE value. Finally, $\bar{u}$ is to satisfy Dirichlet boundary conditions using the SLEDGE output values at α and β .

THEOREM 1 Assume:

(1) the two-point boundary value problem

$$-(p(x)v')' + [q(x) - \lambda r(x)]v = 0$$

on $[\alpha, \beta]$ with $v(\alpha) = v(\beta) = 0$ has only the trivial solution,

(2) $p, q, r, \in C^2[\alpha, \beta]$,

(3) $\epsilon := \max(|\lambda - \bar{\lambda}|, |u(\alpha) - \bar{u}(\alpha)|, |u(\beta) - \bar{u}(\beta)|)$.

Then for ϵ and h sufficiently small $\bar{u}$ exists, and for any $x \in [\alpha, \beta]$

$$u(x) - \bar{u}(x) = O(h^2) + O(\epsilon)$$

and

$$(pu')(x) - (\bar{p}\bar{u}')(x) = O(h^2) + O(\epsilon),$$

where $h = \max_k h_k$.

Proof. Define $e = u - \bar{u}$ and $e' = pu' - \bar{p}\bar{u}'$; it is straightforward to show that

$$\begin{pmatrix} e \\ e' \end{pmatrix}' = \begin{pmatrix} 0 & 1/p \\ q - \lambda r & 0 \end{pmatrix} \begin{pmatrix} e \\ e' \end{pmatrix} + \begin{pmatrix} (\bar{p} - p)\bar{u}'/p \\ (q - \bar{q} - \lambda r + \bar{\lambda}\bar{r})\bar{u} \end{pmatrix}.$$

Let y and z satisfy

$$-(py')' + qy = \lambda ry, \quad y(\alpha) = 1 \quad y(\beta) = 0,$$

and

$$-(pz')' + qz = \lambda rz, \quad z(\alpha) = 0 \quad z(\beta) = 1.$$

With $G(x, t)$ the matrix Green's function for the Sturm-Liouville equation with Dirichlet boundary conditions we have

$$\begin{pmatrix} e(x) \\ e'(x) \end{pmatrix}' = \int_{\alpha}^{\beta} G(x, t) \begin{pmatrix} (\bar{p} - p)\bar{u}'/p \\ (q - \bar{q} - \lambda r + \bar{\lambda}\bar{r})\bar{u} \end{pmatrix} \Big|_t dt \\ + e(\alpha) \begin{pmatrix} y(x) \\ (py')(x) \end{pmatrix} + e(\beta) \begin{pmatrix} z(x) \\ (pz')(x) \end{pmatrix}.$$

The conclusion follows by applying various results from Pruess (1973b). Terms of the form $(q - \bar{q})v$ are shown there to be $O(h^2)$; remaining terms, involving $e(\alpha)$, $e(\beta)$, and $(\lambda - \bar{\lambda})r$, are $O(\epsilon)$. The invertibility of the matrix characterizing $\bar{u}$ follows, for h and ϵ sufficiently small, from assumption (1).

Since the accuracy of $\bar{u}$ as an approximation to u , even with exact data ($\epsilon = 0$), is only $O(h^2)$, we would not expect this to be a very efficient approach except perhaps at modest tolerances. In principle one could use higher order Richardson's extrapolation as SLEDGE does to raise the $O(h^2)$ to $O(h^{2m})$ for some $m > 1$, but this retains the disadvantage of only being effective on a discrete set of output points.

Note that if we required $\bar{p}\bar{u}'$ to interpolate pu' at α and β then it is not difficult to show that the existence of $\bar{u}$ for sufficiently small h would necessarily follow from the uniqueness (up to constant multiples) of eigenfunctions of the Sturm-Liouville problem. In this case assumption (1) of Theorem 1 could be dropped. However, the validity of the assumption can be monitored computationally by checking the condition number (or an estimate) of the associated linear system. If this is too large, switch from Dirichlet interpolation conditions to those of Neumann type.

The problem of being unable to cleanly incorporate all four values at α and β from SLEDGE is inherent to all the methods. For such data, an associated differential equation should be fourth order but (1.1) is only second order.

4 RESIDUAL CORRECTION

It is a simple matter to show that the error $e(x)$ between an exact eigenfunction and an approximation $\bar{u}$ satisfies (for exact data)

$$(pe')' + We = -R(\bar{u}), \quad (4.1)$$

where the residual R is given by

$$R(\bar{u})|_x = (p(x)\bar{u}')' + W(x)\bar{u}.$$

One way to increase the order of accuracy of the SLEDGE approximation is to use residual correction, where we estimate the solution of (4.1) by some numerical technique, e.g., using piecewise polynomial approximations whose parameters are determined by collocation at Gauss points. Equivalently, consider the approximation for $x \in [\zeta_k, \zeta_{k+1}]$

$$(I_2 u)(x) = (I_1 u)(x) + P_d(x), \quad (4.2)$$

where P_d is a piecewise polynomial of degree d or less. We could express P_d on $[\zeta_k, \zeta_{k+1}]$ in monomial form as

$$P_d(x) = \sum_{j=1}^{d+1} \gamma_{kj} \left(\frac{x - \zeta_k}{h_k} \right)^{j-1} \quad (4.3)$$

or use some other representation. We first impose smoothness conditions on P_d ; for the form (4.3), continuity of P_d at ζ_{k+1} gives

$$\sum_{j=1}^{d+1} \gamma_{kj} - \gamma_{k+1,1} = 0$$

while continuity of P'_d there yields

$$\sum_{j=2}^{d+1} (j-1) \gamma_{kj} / h_k - \gamma_{k+1,2} / h_{k+1} = 0.$$

At the endpoints we require $I_2 u = u$ at α and β , i.e., $P_d(\alpha) = P_d(\beta) = 0$. As was noted in the previous section, by ignoring the given pu' values at the ends we lose global continuity of $p(I_2 u)'$. There remain $(d-1)M$ degrees of freedom which are determined by collocation. The error equation (4.1) suggests that we require

$$p[(I_2 - I_1)u]'' + p'[(I_2 - I_1)u]' + W(I_2 - I_1)u \approx -R(I_1 u),$$

i.e., $R(P_d) \approx -R(I_1 u)$. Let ρ_i be the i th zero of the Legendre polynomial of degree $d-1$ scaled to $[0, 1]$, and collocate at

$$\xi_{ki} = \zeta_k + \rho_i h_k. \quad (4.4)$$

In each interval we force

$$R(P_d) = -R(I_1 u)$$

at these $d-1$ Gauss points. This yields

$$\begin{aligned} & W(\xi_{ki}) \gamma_{k1} + \left[\frac{p'(\xi_{ki})}{h_k} + W(\xi_{ki}) \rho_i \right] \gamma_{k2} + \sum_{j=3}^{d+1} \left[\frac{p(\xi_{ki})(j-1)(j-2)}{h_k^2} \right. \\ & \quad \left. + p'(\xi_{ki})(j-1)\rho_i/h_k + W(\xi_{ki})\rho_i^2 \right] \rho_i^{j-3} \gamma_{kj} \\ & = \gamma_{k,d+2} [\tau_k (p(\xi_{ki})\phi'_k + p'(\xi_{ki})\phi_k) - W(\xi_{ki})\phi'_k] \\ & \quad + \gamma_{k,d+3} [\tau_k p(\xi_{ki})\phi_k - p'(\xi_{ki})\phi'_k - W(\xi_{ki})\phi_k]. \end{aligned}$$

The functions ϕ_k and ϕ'_k are evaluated at $\xi_{k,i} - \zeta_k$. We have used $\gamma_{k,d+2}$ for the γ_k of (3.2), and $\gamma_{k,d+3}$ for the δ_k of (3.2); these are known from the first algorithm.

There results a $(d+1)M$ by $(d+1)M$ linear system to be solved with $d > 1$. Note that no collocation conditions are needed for $d = 1$, but then the continuity equations imply that effectively there is just one piece, so we lose the potential increase in accuracy as $h \rightarrow 0$.

The linear system size can be reduced by using a representation for P_d with “built-in” continuity [see Ascher, Pruess, and Russell (1983)]. For example, when $d = 3$ a Hermite representation on $[\zeta_k, \zeta_{k+1}]$ is

$$\begin{aligned} P_3(x) = & U_k \Psi_1 \left(\frac{\zeta_{k+1} - x}{h_k} \right) + U_{k+1} \Psi_1 \left(\frac{x - \zeta_k}{h_k} \right) \\ & - h_k U'_k \Psi_2 \left(\frac{\zeta_{k+1} - x}{h_k} \right) + h_k U'_{k+1} \Psi_2 \left(\frac{x - \zeta_k}{h_k} \right) \end{aligned} \quad (4.5)$$

where $\Psi_1(t) = -2t^3 + 3t^2$ and $\Psi_2(t) = t^3 - t^2$. Then 2 boundary and $2M$ collocation conditions lead to a linear system of order $2M + 2$ in the unknown values U_k and U'_k .

Error bounds (for the case of exact data) follow directly from the theory of de Boor and Swartz (1973).

THEOREM 2 Assume

(1) the two point boundary value problem

$$-(p(x)v')' + [q(x) - \lambda r(x)]v = 0$$

on $[\alpha, \beta]$ with $v(\alpha) = v(\beta) = 0$ has only the trivial solution,

(2) $p, q, r, \in C^2[\alpha, \beta]$.

Then, for exact SLEDGE data, $I_2 u$ exists and for any $x \in [\alpha, \beta]$

$$u(x) - (I_2 u)(x) = O(h^{d+1}),$$

where $h = \max_k h_k$.

There is also superconvergence at the ζ_k if d is sufficiently large.

5 L-SPLINE APPROXIMATIONS

An approach which avoids the computation of I_1 followed by that of I_2 is to start with the representation

$$(I_3 u)(x) = \sum_{j=1}^{d+1} \gamma_{kj} \left(\frac{x - \zeta_k}{h_k} \right)^{j-1} + \gamma_{k,d+2} \phi'_k(x - \zeta_k) + \gamma_{k,d+3} \phi_k(x - \zeta_k). \quad (5.1)$$

where now $\gamma_{k,d+2}$ and $\gamma_{k,d+3}$ are unknown. As before, we force $I_3 u \in C^1[\alpha, \beta]$ and $I_3 u = u$ at α and β . To determine the remaining degrees of freedom we collocate the original differential equation at $d+1$ points per interval, $\{\xi_{ki}\}$, chosen as in (4.4). The continuity restrictions on $I_3 u$ in this monomial form yield

$$\sum_{j=1}^{d+1} \gamma_{kj} + \gamma_{k,d+2} \phi'_k(h_k) + \gamma_{d+3,k} \phi_k(h_k) - \gamma_{k+1,1} - \gamma_{k+1,d+2} = 0,$$

and for $(I_3 u)'$

$$\begin{aligned} \sum_{j=2}^{d+1} \gamma_{kj}(j-1)/h_k - \tau_k \gamma_{k,d+2} \phi_k(h_k) + \gamma_{k,d+3} \phi'_k(h_k) \\ - \gamma_{k+1,2}/h_{k+1} - \gamma_{k+1,d+3} = 0. \end{aligned}$$

Collocation gives

$$\begin{aligned} 0 = & W(\xi_{ki})\gamma_{k1} + [p'(\xi_{ki})/h_k + W(\xi_{ki})\rho_i]\gamma_{k2} \\ & + \sum_{j=3}^{d+1} \left[\frac{p(\xi_{ki})(j-1)(j-2)}{h_k^2} + \frac{p'(\xi_{ki})(j-1)\rho_i}{h_k} \right. \\ & + W(\xi_{ki})\rho_i^2] \rho_i^{j-3} \gamma_{kj} + \gamma_{k,d+2} [-\tau_k (p(\xi_{ki})\phi'_k + p'(\xi_{ki})\phi_k) \\ & \left. + W(\xi_{ki})\phi'_k] + \gamma_{k,d+3} [-\tau_k p(\xi_{ki})\phi_k + p'(\xi_{ki})\phi'_k + W(\xi_{ki})\phi_k]. \end{aligned}$$

The boundary conditions become

$$u(\alpha) = \gamma_{1,1} + \gamma_{1,d+2},$$

and

$$u(\beta) = \sum_{j=1}^{d+1} \gamma_{Mj} + \gamma_{M,d+2} \phi'_M(h_M) + \gamma_{M,d+3} \phi_M(h_M).$$

A $(d+3)M$ by $(d+3)M$ linear system is to be solved.

Note $\bar{u} = I_3 u$ satisfies, with $D = d/dx$,

$$D^{d+1}(p_k D^2 - q_k + \lambda r_k) \bar{u} = 0,$$

i.e., $\bar{u}$ is related to a piecewise L -spline [see Prenter (1971)]. We expect that the convergence rate will be $O(h^{d+3})$ as it is for the L -splines, though the proofs will not be straightforward.

One last possibility should be mentioned, though we have not yet studied it:

$$(I_4 u)(x) := \left[\sum_{j=1}^{d+1} \gamma_{kj} \left(\frac{x - \zeta_k}{h_k} \right)^{j-1} \right] \phi'_k(x - \zeta_k). \quad (5.2)$$

This is more complicated, both algorithmically and mathematically, than the previous ones but it seems to capture the qualitative behavior of $u(x)$ better.

6 NUMERICAL EXAMPLES

Many examples have been tested, results for two are presented here.

Example 1. With $p(x) = 1$, $q(x) = 3/(4x^2)$, $r(x) = 64\pi^2/(9x^6)$ on $[1, 2]$ and Dirichlet boundary conditions, the exact solutions to (1.1) are known, viz.,

$$\lambda_k = (k+1)^2 \quad \text{and} \quad u_k(x) = x^{1.5} \frac{\sqrt{3}}{2\pi} \sin \left[\frac{4(k+1)\pi}{3} \left(1 - \frac{1}{x^2} \right) \right].$$

For the zeroth eigenfunction with SLEDGE data (tolerance 10^{-6} , and $N = 10$ initial mesh points) typical results are found in Table 1. By ‘Method m ’ is meant the results using interpolation scheme I_m . The error ($\|u - \bar{u}\|$ for the max norm) is reduced as expected until the error in the SLEDGE data dominates at $\epsilon = 0.280\text{E} - 8$. As a comparison, standard polynomial spline collocation (no ϕ terms) (with $d = 3$ so error = $O(h^4)$) yields $0.276\text{E} - 4$ when $M = 2$ and $0.760\text{E} - 8$ when $M = 16$. For the zeroth eigenfunction of this simple problem, it probably is the best of all the methods for accuracy and efficiency; however, we would not expect it to do as well for the more oscillatory higher eigenfunctions until the number of ζ intervals is large. While we have not displayed the derivative errors, they are only about ten times larger than the eigenfunction errors even though exact continuity at the points of SLEDGE’s original mesh has been sacrificed. Consequently, SLEDGE’s data values of pu' at mesh points could be used to provide a conservative estimate of the actual error. Table 2 displays the comparable results for the ninth eigenfunction. As we suspected, polynomial collocation does not do as well for this more oscillatory case with an error of 0.480 when $M = 2$ and $0.276\text{E} - 4$ when $M = 16$.

Note: no computing times have been displayed since all data is from just a first crude implementation; however, to achieve a uniform accuracy of 10^{-6} , each interpolation took no more (0.05 to 0.11 seconds) than the clock time of the SLEDGE calculation itself (0.22 seconds).

Table 1. Interpolation error for the zeroth eigenfunction of Example 1 based on SLEDGE data.

M	Method 1	Method 2 (d = 3)	Method 2 (d = 4)	Method 3 (d = 1)	Method 3 (d = 2)
2	0.403E-4	0.849E-5	0.119E-6	0.853E-5	0.113E-6
4	0.720E-5	0.586E-6	0.450E-8	0.588E-6	0.410E-8
8	0.142E-5	0.384E-7	0.280E-8	0.385E-7	0.280E-8
16	0.310E-6	0.310E-8	0.280E-8	0.310E-8	0.280E-8
rate	$O(h^2)$	$O(h^4)$	$O(h^5)$	$O(h^4)$	$O(h^5)$

Table 2. Interpolation error for the ninth eigenfunction of Example 1 based on SLEDGE data.

M	Method 1	Method 2 (d = 3)	Method 2 (d = 4)	Method 3 (d = 1)	Method 3 (d = 2)
2	0.326E-1	0.717E-2	0.225E-2	0.115E-1	0.205E-2
4	0.458E-2	0.639E-3	0.602E-4	0.809E-3	0.436E-4
8	0.996E-3	0.419E-4	0.164E-5	0.472E-4	0.120E-5
16	0.239E-3	0.260E-5	0.410E-6	0.283E-5	0.410E-6
rate	$O(h^2)$	$O(h^4)$	$O(h^5)$	$O(h^4)$	$O(h^5)$

Example 2. With $p(x) = r(x) = 1$,

$$q(x) = \frac{2}{x} - 2000 \left\{ 1 - \left[1 - e^{-1.7(x-1.3)} \right]^2 \right\}$$

on $[0, \infty)$ with Dirichlet boundary conditions, we have a Sturm-Liouville problem for a Morse potential which is singular at both endpoints. At present the interpolation software has only been developed for regular problems, so we apply our methods only on the interior mesh intervals. On the first and last intervals the eigenfunction magnitude is negligible for this example anyway. High accuracy answers can be computed using SLEDGE (or other software) but exact answers are not known here. SLEDGE produced the output data shown in Table 3 for the ninth eigenfunction. Note that the theoretically predicted nine oscillations of $u(x)$ are definitely missed by SLEDGE's initial mesh consisting of just 16 intervals for the tolerance of 10^{-6} . An approximate eigenfunction is plotted in Figure 1 using I_3u with $M = 4$ and $d = 1$. Each SLEDGE data point is indicated with a $\times$. The correct qualitative behavior is evident.

Table 3. SLEDGE data for Example 2.

x	$u(x)$	$(pu')(x)$
0.00000	0.000000	0.000000
0.00044	0.000000	0.000000
0.00246	0.000000	0.000000
0.01382	0.000000	0.000000
0.07773	0.000000	0.000000
0.43714	0.000000	0.000000
0.75334	0.000209	0.012775
0.98420	0.992622	10.524670
1.22340	0.921361	8.309570
1.56162	-0.008026	30.675841
2.02853	-1.419455	-9.243853
2.67263	-0.000282	0.006043
3.56307	0.000000	0.000000
4.78732	0.000000	0.000000
6.47297	0.000000	0.000000
8.80329	0.000000	0.000000

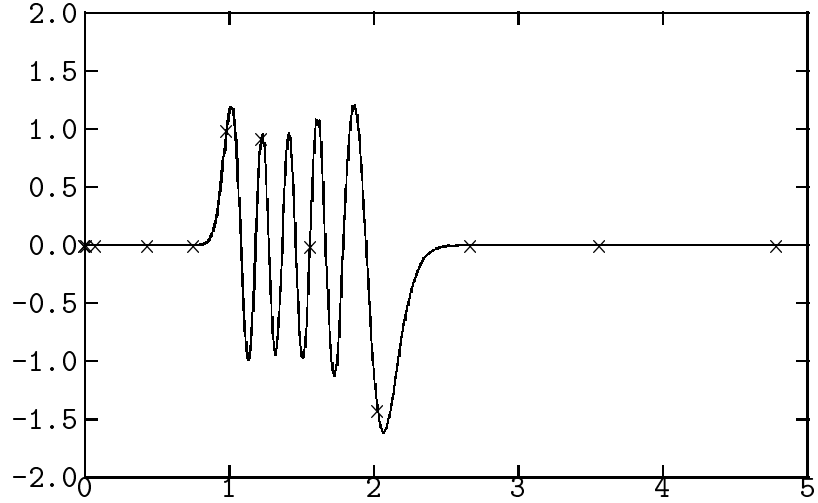


Figure 1. The ninth eigenfunction of a Morse potential (Example 2).

7 CONCLUSIONS

In principle the methods of the previous sections extend to Sturm-Liouville problems with singular endpoints; however, there are some implementation concerns. As noted previously, the infinite interval case requires a change of variable in a neighborhood of $\pm\infty$. Even for a finite singular endpoint, in some cases an asymptotic approximation [see Pruess and Fulton (1995)] must be used to capture the appropriate behavior of the eigenfunction. This is possible from SLEDGE's knowledge of the asymptotic behavior of the coefficient functions. Both of these situations require special treatment algorithmically for our interpolations.

The numerical experiments have confirmed the feasibility of the methods proposed for building high accuracy interpolants from the discrete SLEDGE output. Method 3, especially, looks very attractive because of its efficiency for a given accuracy. More work still needs to be done on (1) the mathematical analysis of Method 3, (2) stability results for the latter two methods, (3) a robust error estimation procedure, and (4) improved implementations of these methods.

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